HITEC University Taxila

Department of Cyber Security

Course: Essentials of Artificial Intelligence

Lab Manual 1

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Lab Tasks

Task 1. Write a Python Code to write the table of a number given by the user.

Input

```
Welcome        lab1 Task 1.py ×
lab1 Task 1.py > ...
1   n = int(input ("enter the number:"))
2   for i in range (1,11):
3       print (n,"*",i,"=",n*i)
```

Output

```
PROBLEMS    OUTPUT    TERMINAL    ...    Python
thon.exe "s:/AI Lab/lab1 Task 1.py"
enter the number:9
9 * 1 = 9
9 * 2 = 18
9 * 3 = 27
9 * 4 = 36
9 * 5 = 45
9 * 6 = 54
9 * 7 = 63
9 * 8 = 72
9 * 9 = 81
9 * 10 = 90
PS S:\AI Lab>
```

Task 2. Write a python code to check wheatear a number is prime or not.

Input

```python
num = int(input("Enter a number: "))

if num <= 1:
    print("Not a prime number")
else:
    is_prime = True

    for i in range(2, num):
        if num % i == 0:
            is_prime = False
            break

    if is_prime:
        print("Prime number")
    else:
        print("Not a prime number")
```

Output

```
PS S:\AI Lab> & C:/Users/HP/AppData/Local/Programs/Python/Python313/py
thon.exe "s:/AI Lab/lab1 Task 2.py"
Enter a number: 7
Prime number
```

Task 3. Write a Python program that prints numbers from 1 to 10 using a while loop.

Input

```python
i = 1

while i <= 10:
    print(i)
    i = i + 1
```

Output

```
PS S:\AI Lab> & C:/Users/HP/AppData/Local/Progra
thon.exe "s:/AI Lab/lab1 Task 3.py"
1
2
3
4
5
6
7
8
9
10
```

Modify the while loop to print only even numbers between 1 and 20.

Input

```
lab1 Task 3.py > ...
i = 2

while i <= 20:
    print(i)
    i = i + 2
```

Output

```
PS S:\AI Lab> & C:/Users/HP/AppData/Local/Programs/
thon.exe "s:/AI Lab/lab1 Task 3.py"
2
4
6
8
10
12
14
16
18
20
```

Task 4. Write a Python program to determine whether the given number is a Harshad Number. If a number is divisible by the sum of its digits, then it will be known as a Harshad Number.

For example:

The number 156 is divisible by the sum (12) of its digits (1, 5, 6) Some Harshad numbers are 54, 120, etc.

Input

```
lab1 Task 4.py > ...
num = int(input("Enter a number: "))

temp = num
digit_sum = 0

while temp > 0:
    digit = temp % 10
    digit_sum = digit_sum + digit
    temp = temp // 10

if num % digit_sum == 0:
    print("Harshad Number")
else:
    print("Not a Harshad Number")
```

Output

```
thon.exe "s:/AI Lab/lab1 Task 4.py"
Enter a number: 120
Harshad Number
```

Conclusion:

This lab successfully demonstrated the application of fundamental Python programming concepts, including loops, conditionals, and arithmetic operations. Each task reinforced logical reasoning and problem-solving skills while highlighting Python's efficiency in handling mathematical and algorithmic problems. Collectively, these exercises provided a strong foundation for advancing toward more complex programming and cybersecurity applications.